

	Covariates	Δelpd	$\text{SE}(\Delta\text{elpd})$	elpd	$\text{SE}(\text{elpd})$	p_{loo}	$\text{se}_{p_{\text{loo}}}$	looic	se_{looic}
IME detect	MLD_obs + MLD_pred + time + month	0.00	0.00	-7304.07	33.48	274.04	3.11	14608.14	66.95
	MLD_obs + time + month	-25.44	7.06	-7329.51	33.11	270.96	3.14	14659.01	66.21
	time + month	-158.82	17.44	-7462.89	27.76	207.13	2.13	14925.78	55.52
IME strength	MLD_obs + MLD_pred + time + month	0.00	0.00	5597.73	76.66	538.42	18.96	-11195.46	153.32
	MLD_obs + time + month	-375.30	33.71	5222.43	74.68	486.53	17.23	-10444.86	149.35
	time + month	-678.29	46.18	4919.44	79.40	459.90	20.99	-9838.88	158.80

Table S1 | Model comparison using leave-out-out cross-validation (LOO). Models fitted to probability of IME detection (IME detect; Bernoulli distributed) and IME strength (chl-a enhancement, %; Gamma-distributed), for nested smoothers of time and month, and including observed mixed layer depth mean and anomaly (MLD_obs), and the long-term MLD trend (MLD_pred). Top-ranked models have the lowest Bayesian LOO of expected log pointwise predictive density (elpd). Additional estimates provided for effective number of parameters (p_{loo}) and LOO information criterion (looic).